

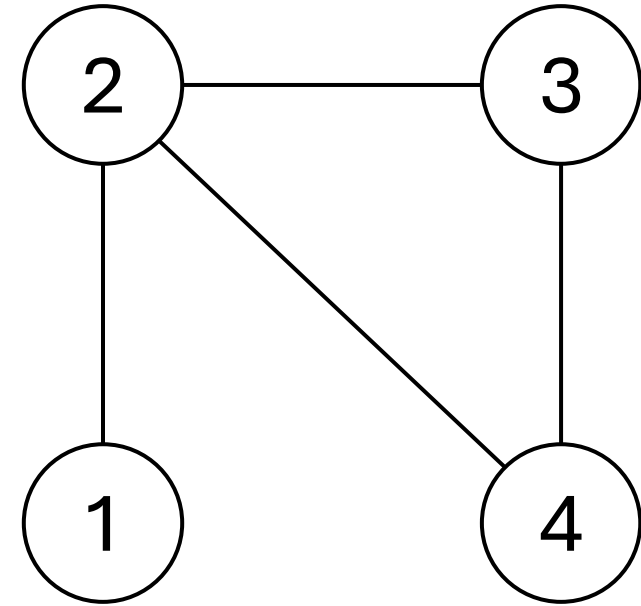


IFORS 2026
The 24th Conference of the International Federation
of Operational Research Societies
12-17 JULY
VIENNA, AUSTRIA

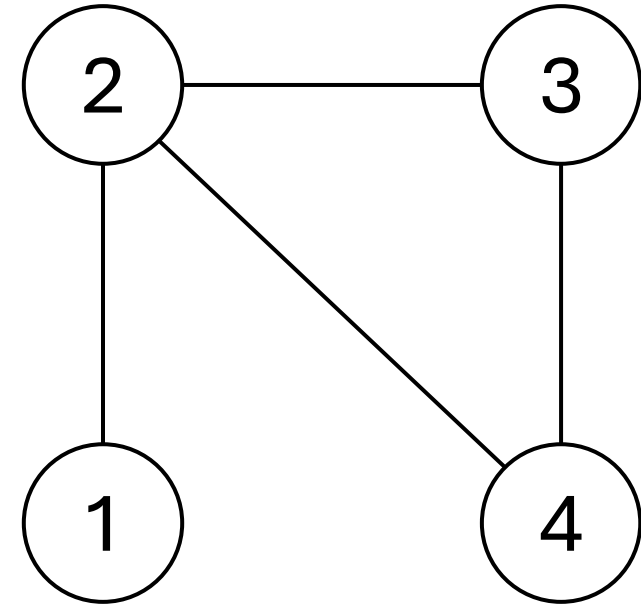
Dynamic matching models under greedy policies: performance paradox and extensions to hypergraphs

Ana Bušić, Josu Doncel, Jean-Michel Fourneau
Shu Li, Doan-Dai Nguyen
INRIA Paris & École Normale Supérieure - PSL

- Motivation: kidney paired donation.
You must have a living donor who is willing and medically able to donate his or her kidney [...]
- Other application: book swap, home swap, etc.
- Features:
 1. Compatibility
 2. Random arrivals
 3. Finite waiting time



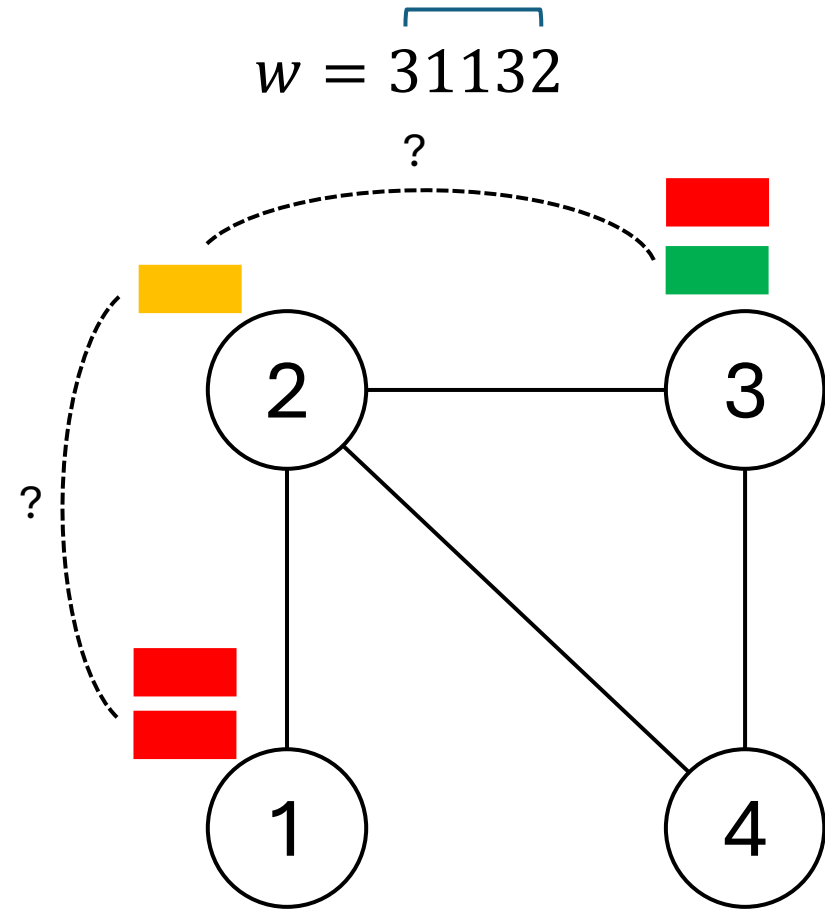
1. Compatibility: An undirected graph $G = (V, E)$
 - Vertices: item classes;
 - Edges: matching types.
If two items of class u and v are in the buffer, and $e = (u, v) \in E$, they *can* form a type- e matching.
 - Matched items leave the buffer.
2. Random arrivals:
 - Distribution μ on V .
 - At time t , a class- v item arrives to the buffer i.i.d. according to μ .
3. Finite waiting time: Stability
 - Under Markovian policies, the buffer is a Markov chain $X(t)$.
 - Stability = X positive recurrent.



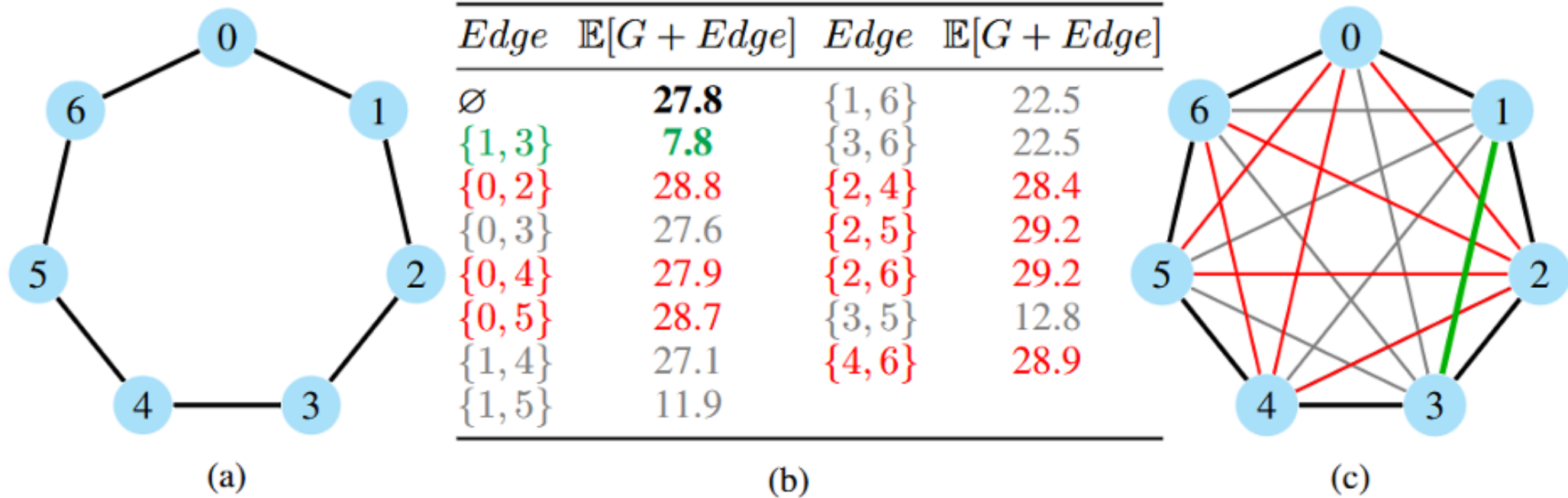
- There can be multiple matchings, and a matching policy is needed.
 - Greedy policies = matching whenever possible
 - Not all policies always stabilise, and not all policies are greedy.
- Example: First-Come-First-Matched (FCFM)
 - Greedy
 - Always stabilising
 - Perceived as fair
- Admit a product-form stationary distribution, but very hard to compute.

$$\pi(w) = \frac{1}{C} \prod_{i=1}^d \frac{\mu(w_i)}{\mu(N(\{w_1, \dots, w_i\}))}$$

$$C = 1 + \sum_{U\text{-independent}} \sum_{\sigma \in \mathcal{S}(|U|)} \prod_{j=1}^{|U|} \frac{\mu(u_{\sigma(j)})}{\mu(N(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})) - \mu(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})}$$



Moyal, P., Bušić, A., and Mairesse, J. (2021). A product form for the general stochastic matching model. Journal of Applied Probability, 58(2):449-468



- Under FCFM policy, performance can worsen if bad edges are added.
- The same phenomenon holds for *all* greedy policies.

Bušić, A., Fourneau, J.-M., Lunven, A., and Li, S. (2025). Performance paradoxes in matching systems are not that rare. In International Conference on Network Games, Artificial Intelligence.

Bušić, A., Cadas, A., Doncel, J., and Fourneau, J.-M. (2024). Performance paradox of dynamic matching models under greedy policies. Queueing Systems, 107(3):257–293.

$$C = 1 + \sum_{U\text{-independent}} \sum_{\sigma \in \mathcal{S}(|U|)} \prod_{j=1}^{|U|} \frac{\mu(u_{\sigma(j)})}{\mu(N(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})) - \mu(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})} > 0?$$

$$\pi(w) = \frac{1}{C} \prod_{i=1}^d \frac{\mu(w_i)}{\mu(N(\{w_1, \dots, w_i\}))}$$

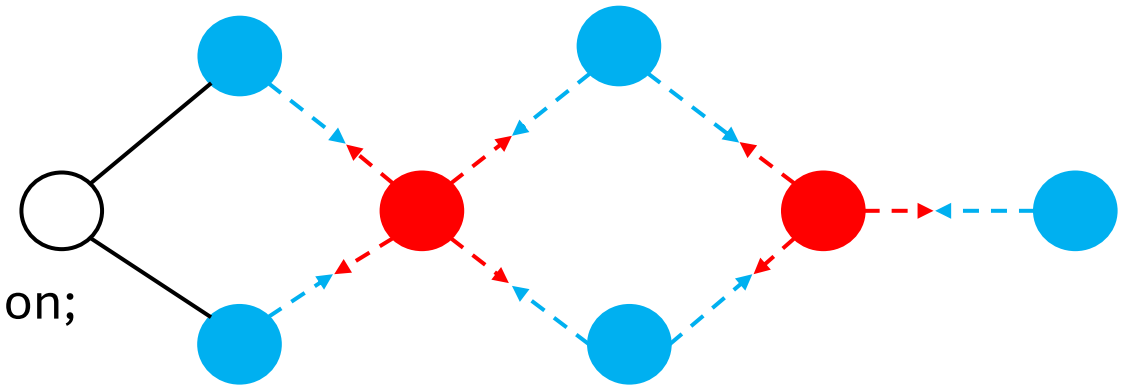
Stability criterion: (G, μ) is stabilizable if **for all independent sets U ,**

$$\mu(U) < \mu(N(U))$$

- Necessity: law of large number
- Sufficiency:
 - negative drift for quadratic Lyapunov function;
 - First-Come-First-Matched, Match-the-Longest.

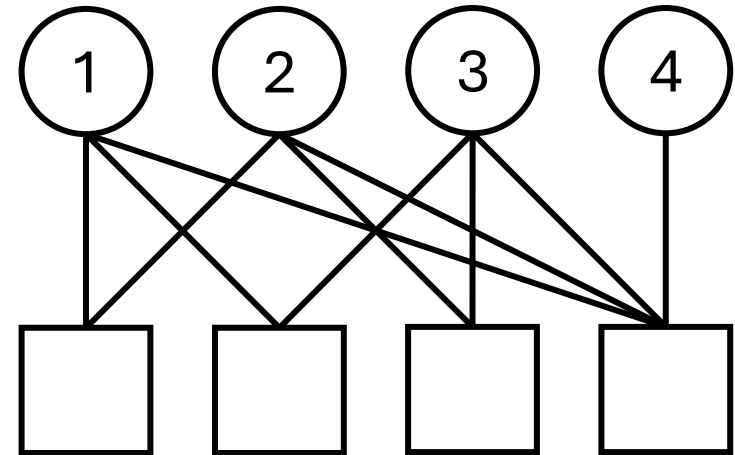
These are greedy policies.

What about *hypergraphs*?



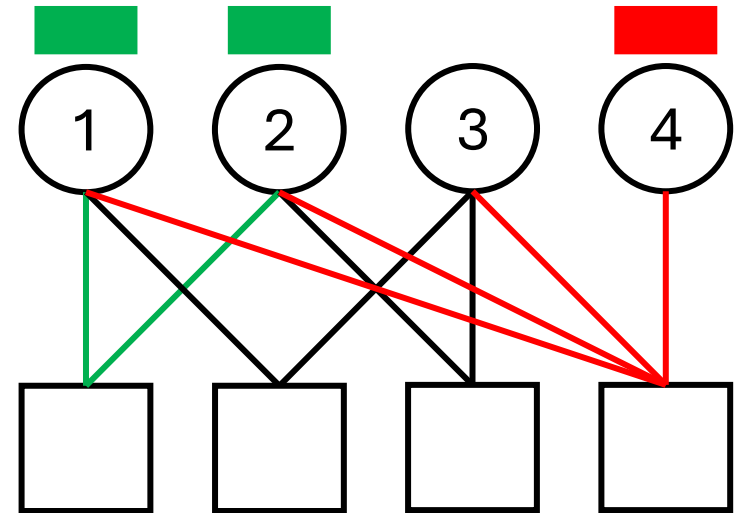
Mairesse, J. and Moyal, P. (2016). Stability of the stochastic matching model. Journal of Applied Probability, 54 (3), pp.1064-1077.

- A **hypergraph** $G = (V, E)$:
 - Vertices: item classes $\bigcirc$;
 - Edges: matching types $\square$.
- Distribution μ on V .
- At time t , a class- v item arrives to the buffer i.i.d. according to μ .
- If **one item for each class u_i** are in the buffer, and $\{u_1, \dots, u_k\} \in E$, they *can* be combined and form a type- e matching.
- Matched items leave the buffer.
- Motivation: multi-party kidney donation.
 - Three-party donation decreases waiting time (Roth, Sönmez, and Ünver, 2007).
- Other applications: multi-stop ride hailing, multi-party good exchange.



Roth, A.E., Sönmez, T., and Utku Ünver, M (2007). Efficient kidney exchange: Coincidence of wants in markets with compatibility-based preferences. *American Economic Review*, 97(3):828-851.

- But greedy policies are *not* maximally stable.
- Problem: given (G, μ) ,
 - What are the necessary and sufficient criteria for stability?
 - If it is stabilisable, find a stabilising policy.
- Question: what is a good alternative to greedy policies?

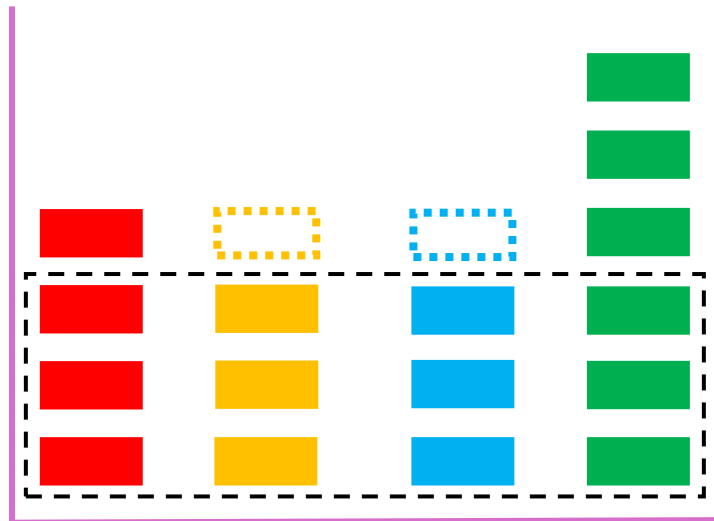
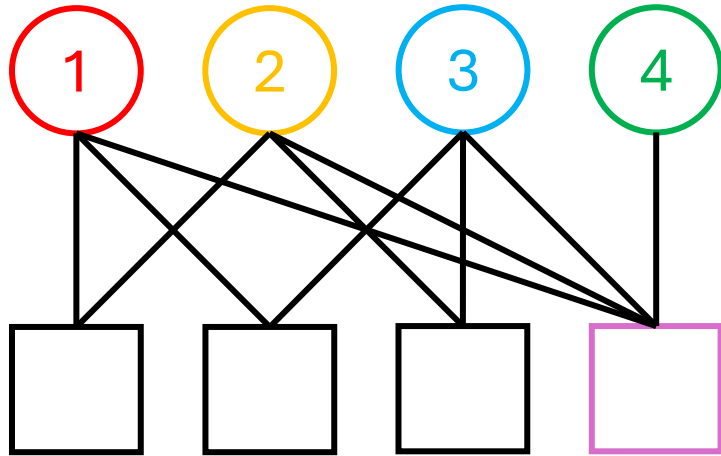


1. Online assignment rate: $\nu = (\nu_{u,e})_{u \in e, e \in E}$
 - $\nu_{u,e}$ is the rate of class- u items being assigned to type- e matchings.
 - Conditions:
 - $\forall e \in E, \forall u \in e, \nu_{u,e} \geq 0$;
 - $\forall u \in V, \sum_{e \ni u} \nu_{u,e} = \mu_u$
2. Online assignment policy:
 - At time t , choose a rate ν .
 - If vertex u is drawn, assign the newly arrived item to type e with probability $\frac{\nu_{u,e}}{\mu_u}$.
 - This item only leaves the buffer via a type- e matching.
3. $U \subset G$ is a support if for all $e \in E$, there exists $v \in e$ such that $(v, e) \notin U$.
 - A buffer state $q = (q_{u,e})_{u \in e, e \in E}$ defines a support $U = \{(u, e) | q_{u,e} > 0\}$.

Stability criterion: (G, μ) is stabilizable iff **for all supports U** , there exists an online assignment rate ν (depending on U) such that for all $(v, e) \in U$,

$$\nu_{v,e} < \frac{1}{|e|} \sum_{u \in e} \nu_{u,e}$$

Nguyen., D.D., and Bušić, A. (2026). Stability in stochastic hypergraph matching: necessity and sufficient criteria. To appear.



Example:

- The support $U \supseteq \{(1, e), (4, e)\}$.
- To have a matching, we need a class-2 and a class-3 item.
- We need them to arrive more often than class-1 and class-4 items to balance out the buffer.
- The stability criterion now reads:

$$\lambda_{\blacksquare} < \frac{\lambda_{\blacksquare} + \lambda_{\blacksquare} + \lambda_{\blacksquare} + \lambda_{\blacksquare}}{4}$$

and

$$\lambda_{\blacksquare} < \frac{\lambda_{\blacksquare} + \lambda_{\blacksquare} + \lambda_{\blacksquare} + \lambda_{\blacksquare}}{4}$$

- How to quantify the imbalance?

$$\begin{aligned} \mathcal{L} = & (\blacksquare - \blacksquare)^2 + (\blacksquare - \blacksquare)^2 + (\blacksquare - \blacksquare)^2 \\ & + (\blacksquare - \blacksquare)^2 + (\blacksquare - \blacksquare)^2 + (\blacksquare - \blacksquare)^2 \end{aligned}$$

Sketch of proof:

- **Sufficiency:** let $X(t) = q$,

- Lyapunov function: $\mathcal{L}(q) = \sum_{e \in E} \sum_{\{u,v\} \in e} (q_{u,e} - q_{v,e})^2$.
- Given an assignment rate ν for some constant C_0

$$\mathbb{E}[\Delta \mathcal{L} | X(t) = q] \leq C_0 + \sum_{(v,e) \in U} q_{v,e} |e| \left(\nu_{v,e} - \frac{1}{|e|} \sum_{u \in e} \nu_{u,e} \right) \leq -\varepsilon$$

- Stability criterion gives negative drift for all $\|q\|_1$ large enough.

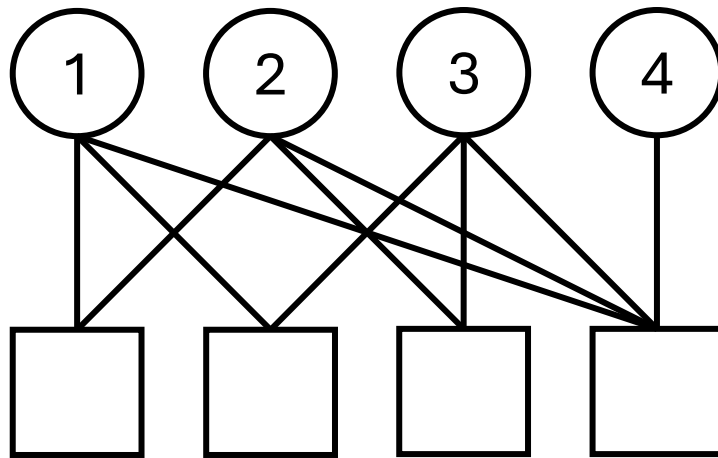
- **Necessity:** suppose the criterion fails for a support U ,

- Define $Y(t) = (Y_{u,e}(t))_{(u,e) \in U}$ by

$$Y_{v,e}(t) = X_{v,e}(t) - \frac{1}{|e|} \sum_{u \in e} X_{u,e}(t)$$

- X is a Markov chain, thus so is Y
- $X(\tau) = 0 \Rightarrow Y(\tau) = 0$.
- (G, μ) stabilisable $\Leftrightarrow \mathbb{E}[\tau] < \infty \Rightarrow Y$ positive recurrent.
- But the criterion fails, Y cannot be positive recurrent.

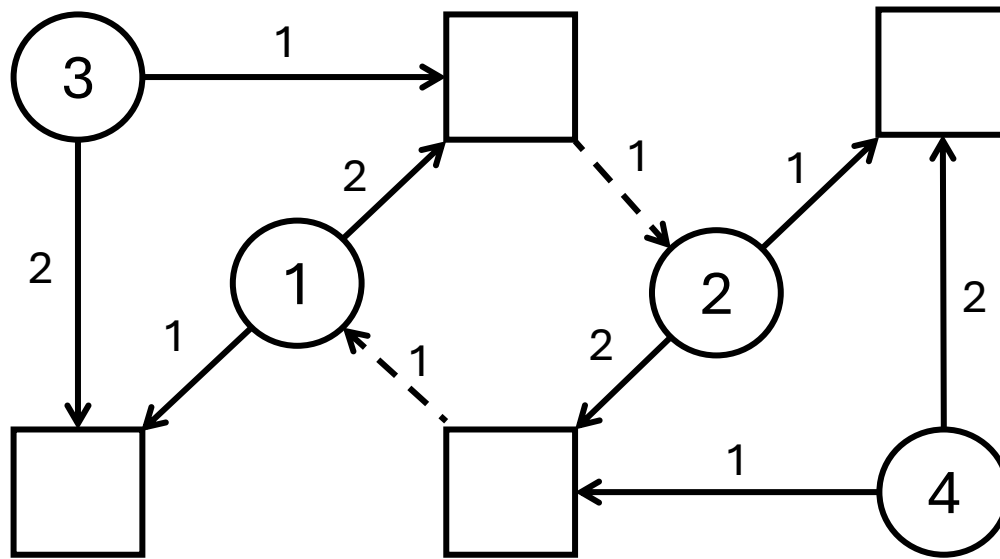
- Let $A \in \{0, 1\}^{V \times E}$ be the incidence matrix of G . $A_{u,e} = 1$ iff $u \in e$.
- Stability criteria using A :
 - (Comte et al., 2021) When G is a graph, (G, μ) is stabilisable iff
 - A is surjective, and
 - There exists $\lambda \in \mathbb{R}_{\geq 0}^E$ such that $A\lambda = \mu$.
 - (N., Bušić, 2026) The same holds when G is a hypergraph.
- Consequences: stability is verifiable in polynomial time.



$$A = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \det A = 2$$

Comte, C., Mathieu, F., Varma, S. M., and Bušić, A. (2025). Online Stochastic Matching: A Polytope Perspective. arXiv:2112.14457
 Nguyen, D.D., and Bušić, A. (2026). Stability in stochastic hypergraph matching: necessary and sufficient criteria. To appear.

- Some generalisations:
 - Non-Markovian policies;
 - Reinterpreting A as matching matrix;
 - Abandonment, service time;
 - Probabilistic matching.
- Applications:
 - Matching with multiplicities;
 - Routing, production line.



$$A = \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 2 \\ 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \end{pmatrix}$$

“Greedy policies are bad.”
- Georgios Paschos